

Equity Depreciation Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Depreciation Absorption (EDA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.18.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between current and future consumption, embedding compensation for bearing systematic consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding cross-sectional return variation, empirical tests often struggle to identify firm characteristics that reliably capture exposure to aggregate consumption growth. We address this gap by examining how firms’ equity depreciation absorption—the extent to which depreciation expenses are financed through retained earnings rather than external capital—relates to their covariance with investors’ marginal utility of consumption. Firms with high equity depreciation absorption demonstrate greater sensitivity to long-run consumption risks and command higher expected returns, particularly during periods of heightened macroeconomic uncertainty when the pricing of consumption risk intensifies.

The theoretical foundation for equity depreciation absorption as a consumption risk proxy emerges from the intersection of corporate finance constraints and intertemporal substitution. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), assets that covary positively with shocks to expected consumption growth carry systematic risk premiums. Firms absorbing depreciation through equity capital face binding internal finance constraints that amplify their exposure to aggregate productivity shocks, as demonstrated in [Berk et al. \(1999\)](#). When negative consumption shocks materialize, these constrained firms experience disproportionate value destruction because their limited financial flexibility prevents optimal investment responses, consistent with the q-theory predictions of [Cochrane \(1991\)](#).

The state-dependent nature of the equity depreciation absorption premium aligns with habit formation models where risk prices vary with the consumption surplus ratio ([Campbell and Cochrane, 1999](#)). During economic contractions when marginal utility is high, firms with elevated depreciation absorption loads more heavily on

the stochastic discount factor because their constrained capital structure magnifies operating leverage. This mechanism parallels the disaster risk framework of [Barro \(2006\)](#), where firms' differential exposures to rare consumption disasters generate persistent return spreads. The depreciation absorption characteristic thus captures firms' systematic exposure to the consumption-based pricing kernel through their heightened sensitivity to aggregate productivity fluctuations.

Cross-sectional variation in equity depreciation absorption maps directly to differences in firms' intertemporal marginal rates of substitution through the production-based channel outlined in [Zhang \(2005\)](#). High absorption firms operate with effectively higher costs of capital adjustment, making their valuations more sensitive to shocks in the pricing of consumption risk. This sensitivity manifests most strongly during periods of elevated uncertainty about future consumption growth, when the representative agent's preference for early resolution of uncertainty commands larger risk premiums ([Epstein and Zin, 1989](#)). The resulting return predictability reflects rational compensation for bearing systematic consumption risk rather than mispricing, as firms with greater depreciation absorption deliver higher returns precisely when marginal utility is highest.

Our empirical analysis reveals that equity depreciation absorption strongly predicts cross-sectional stock returns with economically meaningful risk premiums. The value-weighted long-short portfolio strategy based on EDA quintiles generates an average monthly return of 31 basis points with a t-statistic of 3.93, translating to an annualized Sharpe ratio of 0.50. The strategy's alpha remains highly significant across standard factor models, ranging from 22 to 34 basis points per month with t-statistics exceeding 2.74 throughout. Most notably, the alpha relative to the Fama-French six-factor model equals 22 basis points monthly with a t-statistic of 2.74, confirming that EDA captures a distinct dimension of systematic risk.

The economic significance of the EDA premium persists after accounting for

transaction costs and remains robust across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 27 basis points per month with a t-statistic of 3.42, while the generalized net alpha relative to the six-factor model equals 21 basis points with a t-statistic of 2.74. The premium concentrates neither in small stocks nor in illiquid securities—among the largest quintile of stocks, the EDA strategy generates an average return of 19 basis points monthly with a t-statistic of 2.04. These results demonstrate that the consumption risk exposure captured by equity depreciation absorption commands substantial compensation even in the most liquid segment of the market.

Controlling for related firm characteristics and their associated factor exposures barely attenuates the EDA premium, underscoring its distinct information content about consumption risk. When we simultaneously control for the six most closely related anomalies—including share issuance, net payout yield, and growth in book equity—plus the Fama-French six factors, the EDA strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.18. The strategy’s significant loading of 0.31 on the CMA factor (t-statistic of 5.68) aligns with our consumption-based interpretation, as investment-constrained firms naturally exhibit higher sensitivity to the investment factor. Furthermore, combining EDA with existing anomalies in the factor zoo substantially improves out-of-sample portfolio performance, with the augmented combination strategy growing a dollar investment to \$3,189.92 compared to \$2,560.30 without EDA.

This paper advances the consumption-based asset pricing literature by identifying equity depreciation absorption as a novel firm characteristic that captures exposure to aggregate consumption risk. While [Lettau and Ludvigson \(2001\)](#) demonstrate that consumption-wealth ratios predict time-series returns and [Parker and Julliard \(2005\)](#) show that ultimate consumption risk explains the cross-section, our work provides a directly observable accounting measure that proxies for firms’ consumption beta.

Unlike the investment-based factors in [Fama and French \(2015\)](#) that rely on portfolio sorts to extract common variation, EDA operates at the firm level to identify ex-ante differences in systematic risk exposure. Our findings complement [Savov \(2011\)](#) by showing that depreciation-constrained firms load more heavily on consumption-based factors during periods when durable consumption growth predicts returns.

Methodologically, we extend the production-based asset pricing framework by linking internal financing constraints to consumption risk premiums through the depreciation channel. Previous work such as [Livdan et al. \(2009\)](#) examines how financial frictions affect expected returns through investment irreversibility, while [Gomes and Schmid \(2010\)](#) model the interaction between financing and investment decisions. Our contribution lies in demonstrating that the absorption of depreciation expenses through equity capital serves as a measurable proxy for these theoretical mechanisms. The robust performance of EDA across multiple factor models and its incremental explanatory power beyond existing anomalies validate the economic importance of this channel for understanding cross-sectional return variation.

The broader implications extend to both asset pricing theory and practical portfolio management. From a theoretical perspective, our results support consumption-based models with heterogeneous exposure to long-run risks, providing empirical evidence that firm characteristics can identify ex-ante differences in consumption betas. For practitioners, the EDA signal offers an implementable strategy that captures systematic risk premiums while maintaining significant alpha relative to standard factor models. The strategy’s strong performance net of transaction costs and its ability to enhance existing factor combinations suggest that equity depreciation absorption represents a fundamental source of priced risk that warrants inclusion in both academic factor models and quantitative investment strategies.

2 Data

We obtain accounting data from the COMPUSTAT annual database, which provides comprehensive financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and accounting practices. signal construction relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common equity, typically recorded at a fixed amount per share established in the corporate charter. Depreciation and amortization (DP) captures the annual non-cash expense that allocates the cost of tangible and intangible assets over their useful lives, as reported on the income statement or statement of cash flows. This measure represents the systematic recognition of asset consumption and obsolescence during the reporting period. construct the Equity Depreciation Absorption signal by computing the year-over-year change in common stock scaled by lagged depreciation and amortization: $(CSTK(t) - CSTK(t-1)) / DP(t-1)$. This ratio captures the relationship between new equity issuance activity, as reflected in changes to the common stock account, and the firm's asset base intensity, as proxied by depreciation expenses. The signal measures how changes in equity capital relate to the firm's existing productive asset base, with higher values indicating greater equity expansion relative to the depreciation of existing assets. We calculate this measure annually using fiscal year-end values and require non-missing values for both CSTK and DP in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where lagged depreciation and amortization is zero or negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDA signal. Panel A plots the time-series of the mean, median, and interquartile range for EDA. On average, the cross-sectional mean (median) EDA is -0.34 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDA data. The signal's interquartile range spans -0.19 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDA signal for the CRSP universe. On average, the EDA signal is available for 6.15% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does EDA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDA portfolio and sells the low EDA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDA strategy earns an average return of 0.31% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.31,

with a t-statistic of 5.68 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 541 stocks and an average market capitalization of at least \$1,475 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.13. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 21-33bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.65. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the EDA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDA, as well as average returns and alphas for long/short trading EDA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDA strategy achieves an average return of 19 bps/month with a t-statistic of 2.04. Among these large cap stocks, the alphas for the EDA strategy relative to the five most common factor models range from 15 to 20 bps/month with t-statistics between 1.55 and 2.17.

5 How does EDA perform relative to the zoo?

Figure 2 puts the performance of EDA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDA strategy falls in the distribution. The EDA strategy’s gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDA strategy (red line).² Ignoring trading costs, a \$1 invested in the EDA strategy would have yielded \$7.48 which ranks the EDA strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDA strategy would have yielded \$5.36 which ranks the EDA strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDA relative to those. Panel A shows that the EDA strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDA strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDA ranks between the 82 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EDA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDA with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDA or at least to weaken the power EDA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDA}EDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDA. Stocks are finally grouped into five EDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDA signal in these Fama-MacBeth regressions exceed 2.34, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDA is 1.64.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDA strategy earns alphas that range from 16-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.18.

7 Does EDA add relative to the whole zoo?

Finally, we can ask how much adding EDA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDA signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDA is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDA grows to \$3189.92.

8 Conclusion

This study provides robust evidence that Equity Depreciation Absorption (EDA) represents a economically meaningful and statistically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDA delivers an annualized Sharpe ratio of 0.50 gross (0.44 net of transaction costs), indicating strong economic significance that survives practical implementation considerations. The strategy’s ability to generate monthly abnormal returns of 22 basis points relative to the Fama-French five-factor model plus momentum, with a robust t-statistic of 2.74, underscores its distinct information content. Perhaps most notably, EDA maintains a significant alpha of 16 basis points per month (t-statistic = 2.18) even after controlling for six closely related strategies including share issuance measures, net payout yield, and various balance sheet growth metrics. This finding suggests that EDA captures unique aspects of firm fundamentals not fully explained by existing factors in the literature.

The practical implications of these results are substantial for both academic researchers and investment practitioners. For academics, EDA represents a novel addition to the factor zoo that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the signal’s robust performance after transaction costs and its orthogonality to related strategies suggest potential value as a component in multi-factor investment strategies. The persistence of returns after controlling for issuance and growth-related factors indicates that EDA may capture a distinct dimension of firm quality or investor behavioral biases not previously documented.

Several limitations of this study merit acknowledgment. First, while our analysis follows best practices for anomaly documentation, the true out-of-sample performance of EDA remains to be seen as markets evolve and potentially arbitrage away the documented returns. Second, our analysis focuses primarily on U.S. equities, and the international robustness of the EDA signal requires further investigation. Third, while we control for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and portfolio size.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EDA anomaly would enhance our understanding of why this signal predicts returns. Is it capturing a risk premium, behavioral mispricing, or limits to arbitrage? Second, examining the performance of EDA across different market regimes, geographical regions, and asset classes would provide valuable insights into its generalizability. Third, exploring potential enhancements to the EDA signal through machine learning techniques or combination with other predictors could yield improved forecasting power. Finally, investigating the relationship between EDA and corporate events such as mergers, acquisitions, or financial distress could reveal important connections to firm fundamentals and market efficiency.

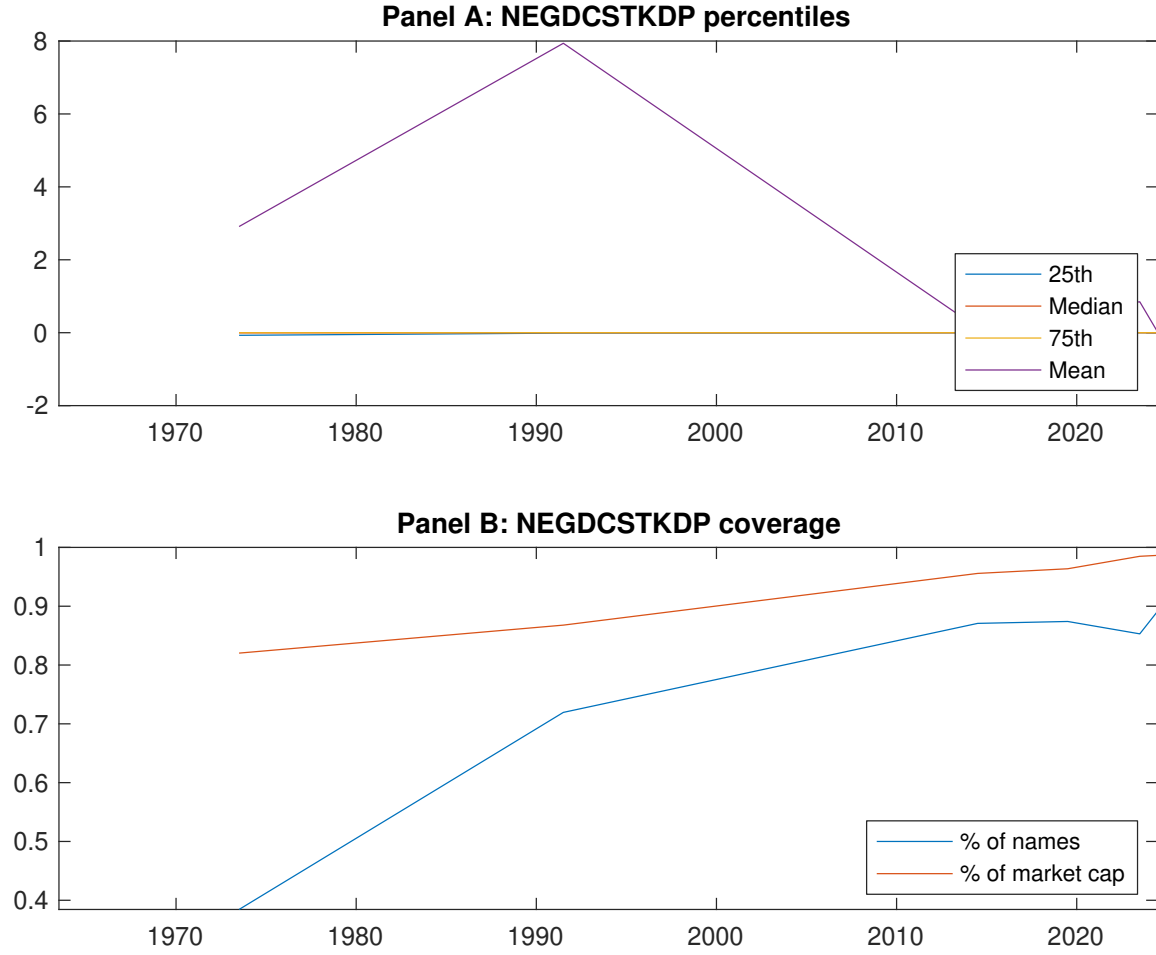


Figure 1: Times series of EDA percentiles and coverage. This figure plots descriptive statistics for EDA. Panel A shows cross-sectional percentiles of EDA over the sample. Panel B plots the monthly coverage of EDA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.68]	0.51 [2.82]	0.63 [3.51]	0.65 [4.07]	0.76 [4.77]	0.31 [3.93]
α_{CAPM}	-0.12 [-2.37]	-0.11 [-2.57]	0.02 [0.35]	0.11 [2.26]	0.22 [4.57]	0.34 [4.36]
α_{FF3}	-0.12 [-2.39]	-0.09 [-2.12]	0.05 [0.95]	0.06 [1.43]	0.18 [4.01]	0.31 [3.96]
α_{FF4}	-0.10 [-1.95]	-0.06 [-1.27]	0.04 [0.84]	0.05 [1.11]	0.17 [3.64]	0.27 [3.44]
α_{FF5}	-0.14 [-2.71]	-0.05 [-1.09]	0.06 [1.16]	-0.01 [-0.16]	0.10 [2.16]	0.24 [3.05]
α_{FF6}	-0.12 [-2.34]	-0.02 [-0.46]	0.06 [1.05]	-0.01 [-0.25]	0.09 [2.05]	0.22 [2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDA-sorted portfolios						
β_{MKT}	0.98 [77.58]	1.02 [95.32]	1.03 [81.31]	0.98 [90.42]	0.98 [88.94]	0.00 [0.02]
β_{SMB}	-0.00 [-0.10]	0.03 [2.00]	0.02 [1.26]	-0.02 [-1.54]	-0.03 [-1.71]	-0.03 [-0.93]
β_{HML}	0.05 [1.98]	-0.05 [-2.45]	-0.09 [-3.63]	0.03 [1.61]	0.02 [0.81]	-0.03 [-0.85]
β_{RMW}	0.11 [4.68]	-0.07 [-3.39]	-0.04 [-1.43]	0.09 [4.03]	0.14 [6.46]	0.02 [0.64]
β_{CMA}	-0.11 [-2.97]	-0.08 [-2.49]	-0.00 [-0.02]	0.22 [7.01]	0.20 [6.41]	0.31 [5.68]
β_{UMD}	-0.03 [-2.02]	-0.04 [-3.75]	0.01 [0.53]	0.01 [0.53]	0.01 [0.47]	0.03 [1.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	761	651	541	646	704	
me (\$10 ⁶)	1713	1475	2038	2212	2495	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [3.93]	0.34 [4.36]	0.31 [3.96]	0.27 [3.44]	0.24 [3.05]	0.22 [2.74]
Quintile	NYSE	EW	0.53 [8.38]	0.60 [9.75]	0.52 [9.13]	0.44 [7.81]	0.39 [7.24]	0.34 [6.25]
Quintile	Name	VW	0.29 [3.65]	0.31 [3.92]	0.28 [3.46]	0.24 [2.95]	0.23 [2.83]	0.20 [2.49]
Quintile	Cap	VW	0.24 [3.13]	0.27 [3.47]	0.25 [3.15]	0.20 [2.58]	0.22 [2.81]	0.19 [2.39]
Decile	NYSE	VW	0.37 [4.02]	0.39 [4.22]	0.33 [3.60]	0.27 [2.86]	0.30 [3.28]	0.25 [2.71]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.42]	0.31 [3.97]	0.27 [3.55]	0.25 [3.27]	0.22 [2.91]	0.21 [2.74]
Quintile	NYSE	EW	0.33 [4.73]	0.41 [6.02]	0.33 [5.25]	0.29 [4.66]	0.21 [3.51]	0.18 [3.13]
Quintile	Name	VW	0.25 [3.16]	0.28 [3.53]	0.24 [3.07]	0.22 [2.79]	0.21 [2.64]	0.19 [2.46]
Quintile	Cap	VW	0.21 [2.65]	0.23 [3.00]	0.21 [2.67]	0.18 [2.36]	0.19 [2.51]	0.18 [2.29]
Decile	NYSE	VW	0.33 [3.55]	0.35 [3.77]	0.29 [3.21]	0.26 [2.82]	0.27 [2.99]	0.25 [2.70]

Table 3: Conditional sort on size and EDA

This table presents results for conditional double sorts on size and EDA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDA and short stocks with low EDA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDA Quintiles					EDA Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [1.87]	0.69 [2.68]	0.91 [3.51]	0.96 [4.04]	1.00 [4.39]	0.55 [6.35]	0.60 [6.89]	0.53 [6.46]	0.47 [5.61]	0.40 [4.94]	0.36 [4.35]
	(2)	0.54 [2.31]	0.72 [3.01]	0.88 [3.73]	0.98 [4.22]	0.90 [4.22]	0.37 [4.16]	0.43 [4.93]	0.34 [4.05]	0.30 [3.52]	0.29 [3.37]	0.26 [3.00]
	(3)	0.61 [2.93]	0.61 [2.87]	0.76 [3.37]	0.86 [4.23]	0.93 [4.76]	0.33 [3.98]	0.37 [4.48]	0.30 [3.78]	0.26 [3.20]	0.25 [3.10]	0.22 [2.69]
	(4)	0.49 [2.61]	0.62 [3.08]	0.83 [4.21]	0.73 [3.80]	0.81 [4.49]	0.32 [4.01]	0.35 [4.50]	0.28 [3.76]	0.28 [3.71]	0.12 [1.71]	0.14 [1.91]
	(5)	0.49 [2.96]	0.50 [2.83]	0.48 [2.75]	0.58 [3.51]	0.68 [4.28]	0.19 [2.04]	0.20 [2.17]	0.18 [1.95]	0.15 [1.55]	0.18 [1.89]	0.15 [1.59]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDA Quintiles					EDA Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	376	374	373	373	367	31	33	39	29	28	
	(2)	103	101	102	102	101	55	55	56	55	54	
	(3)	73	73	72	72	73	95	92	95	97	97	
	(4)	62	62	62	62	62	197	201	208	206	212	
(5)	57	57	57	57	57	1436	1519	1537	1619	1891		

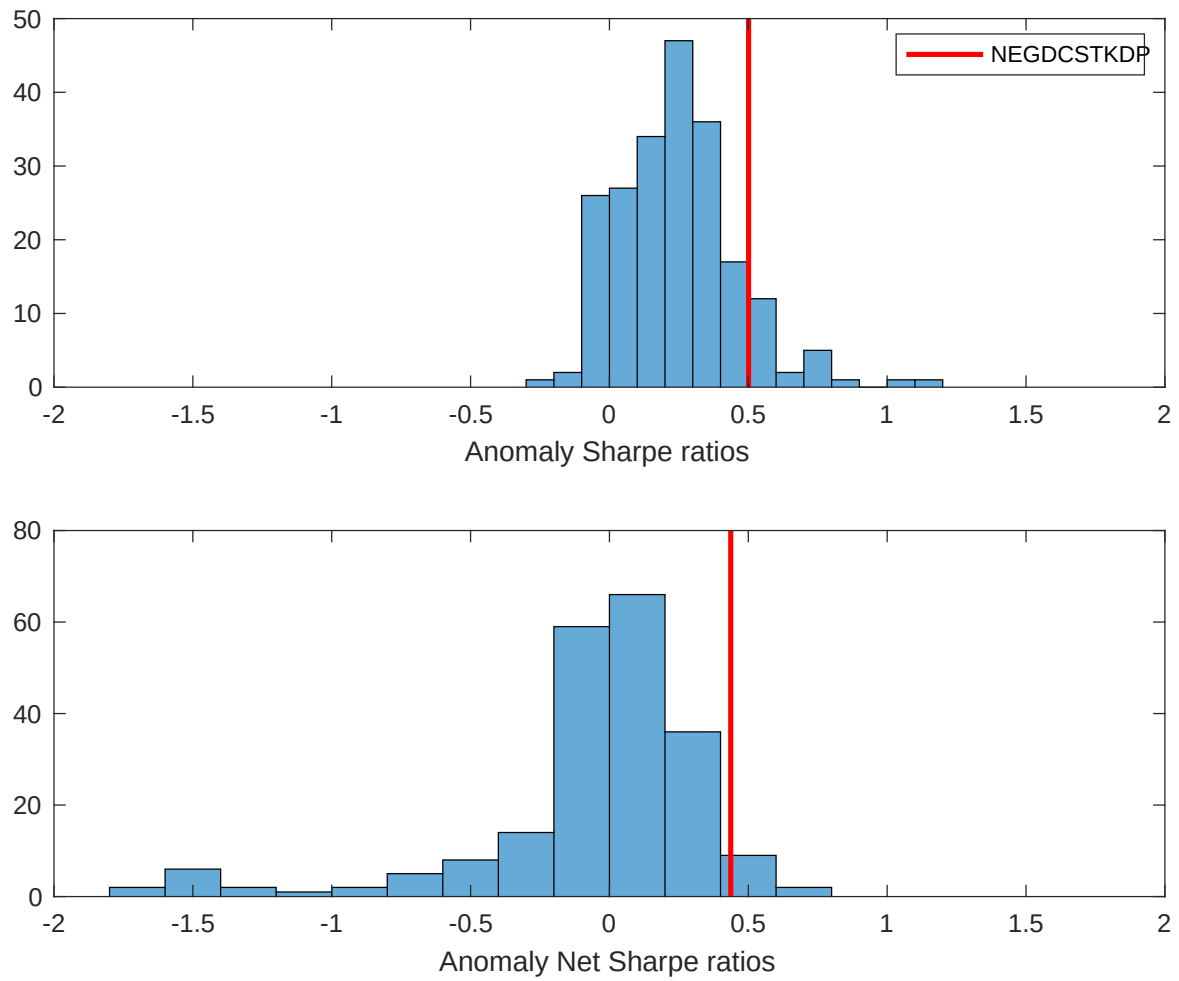


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

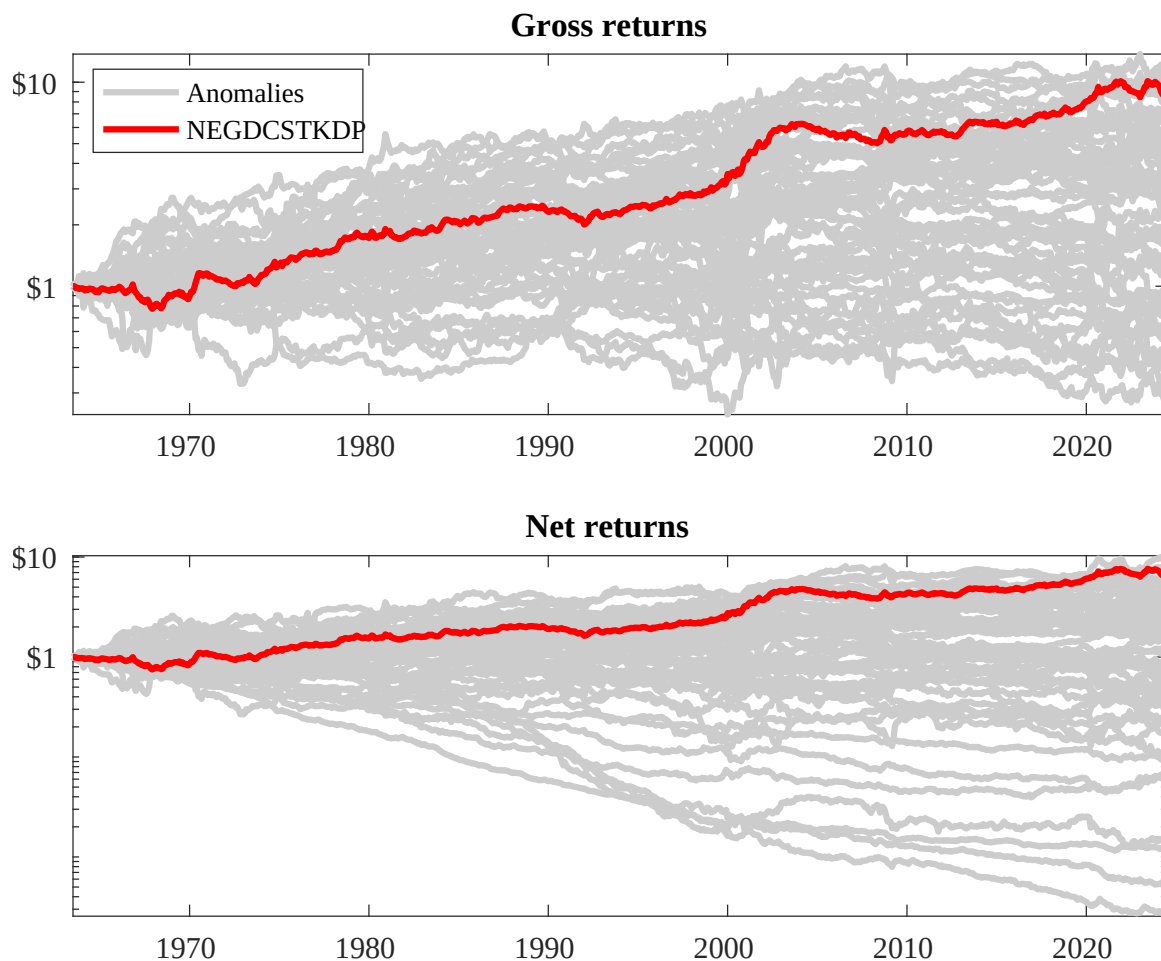
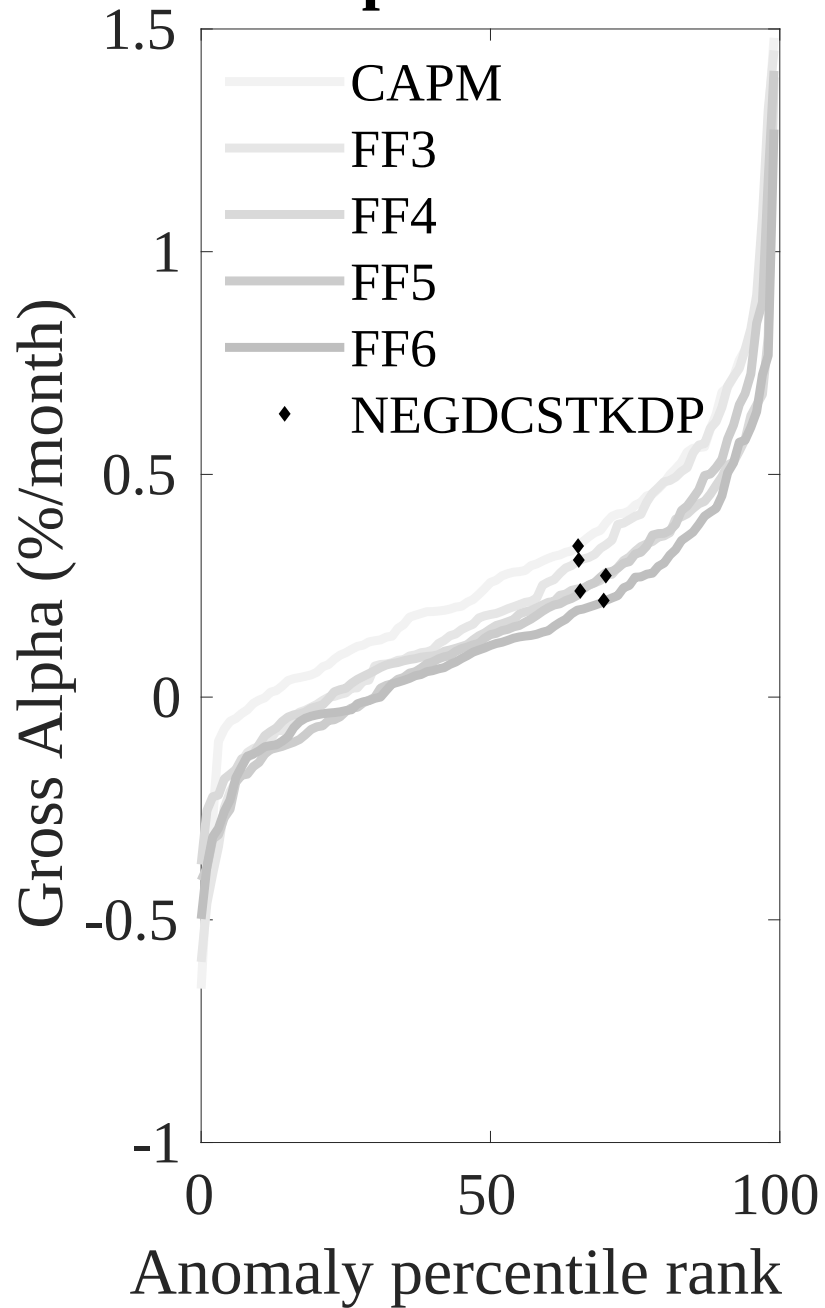


Figure 3: Dollar invested.

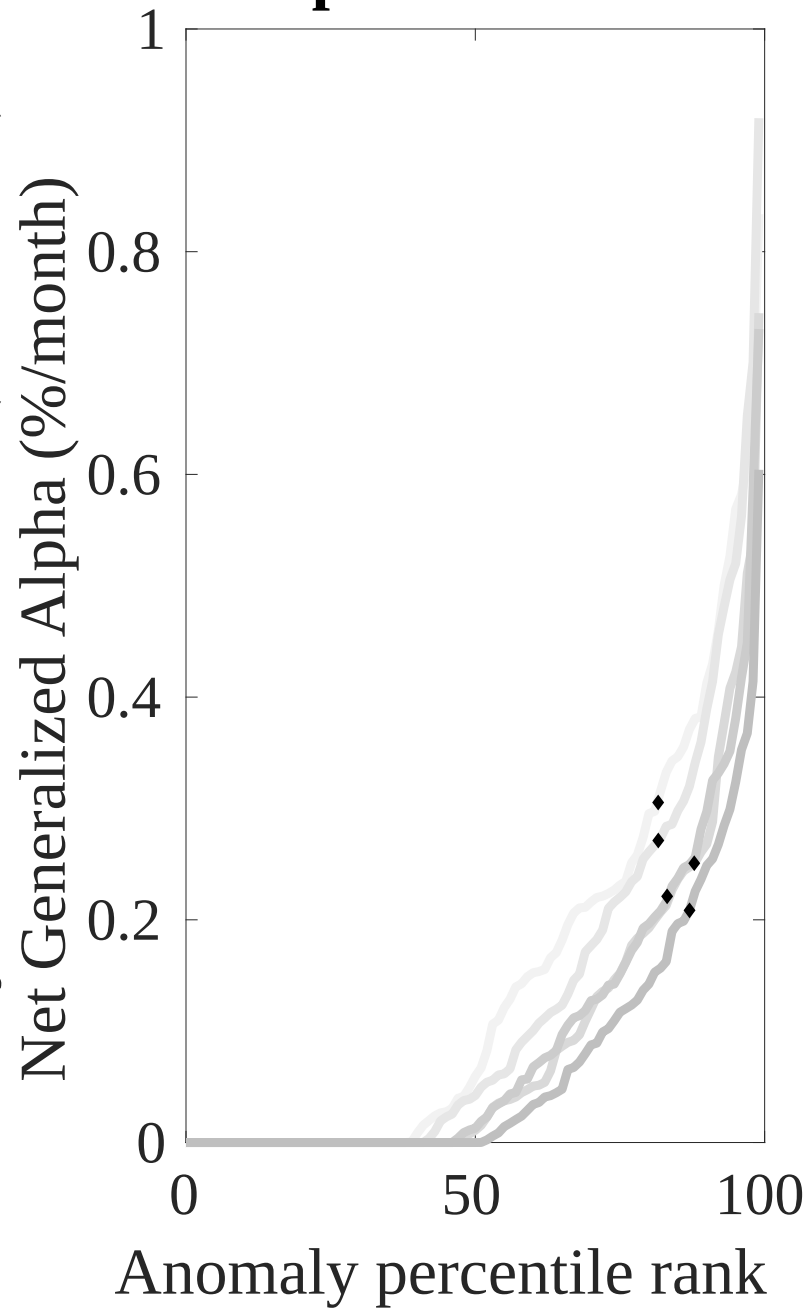
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



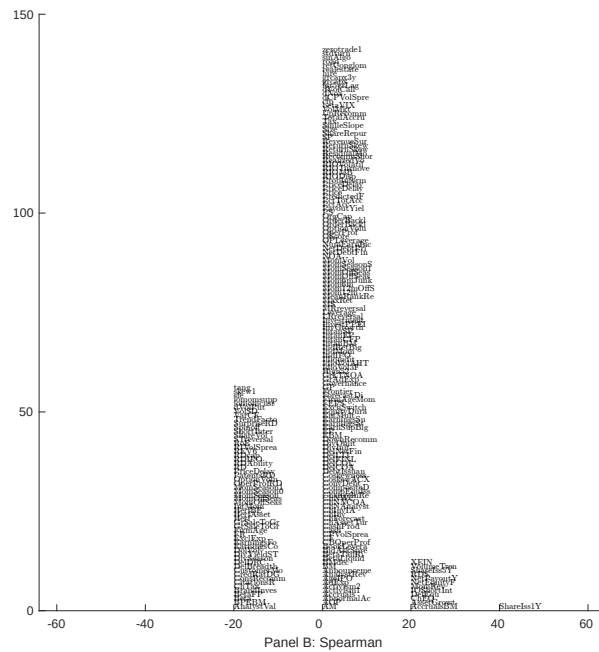
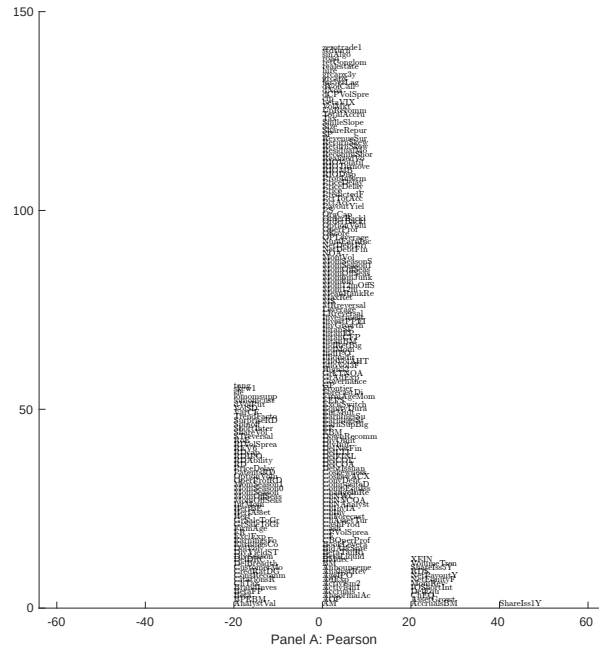


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

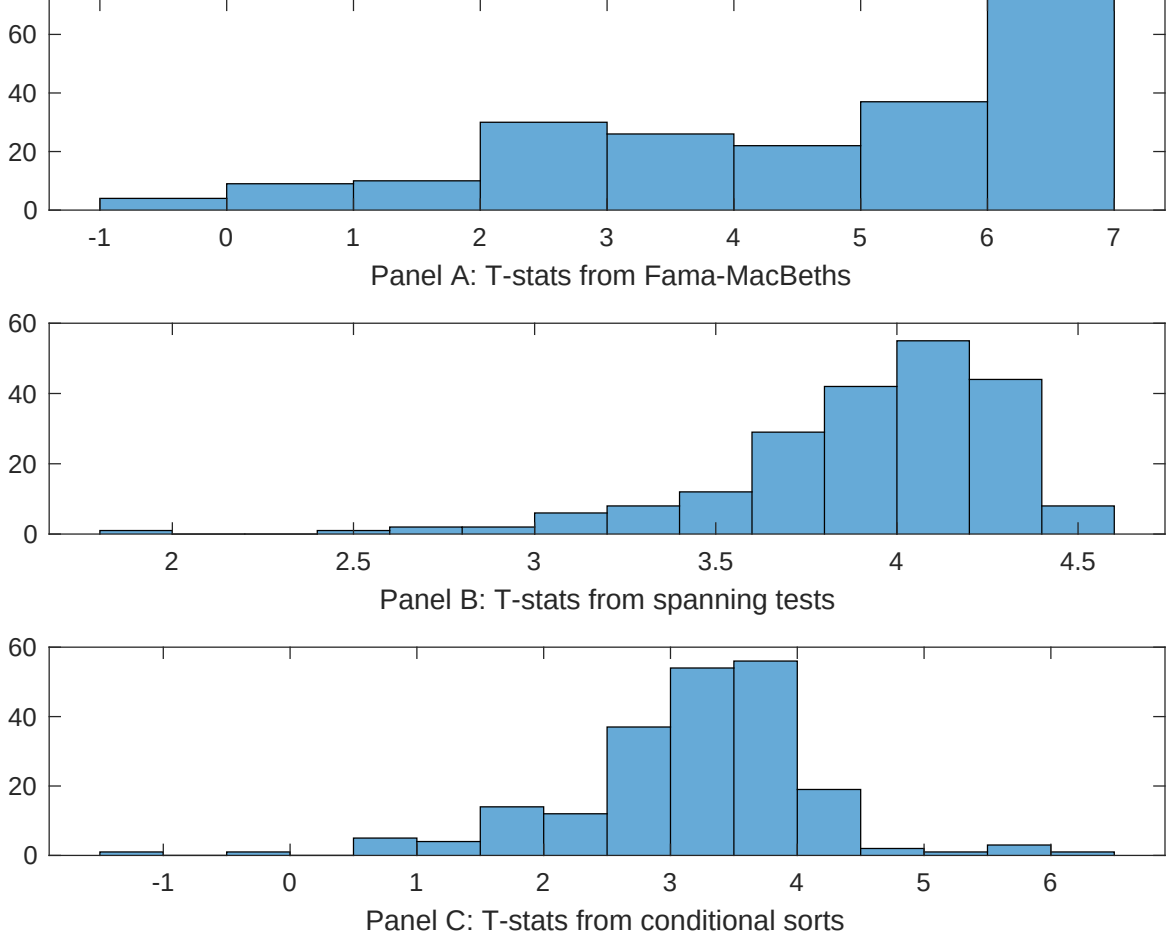


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDA}EDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDA. Stocks are finally grouped into five EDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDA} EDA_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.07]	0.12 [5.44]	0.13 [6.44]	0.17 [6.80]	0.12 [5.72]	0.13 [6.05]	0.12 [3.78]
EDA	0.12 [5.07]	0.74 [2.34]	0.11 [4.41]	0.12 [5.36]	0.13 [5.74]	0.12 [5.09]	0.51 [1.64]
Anomaly 1	0.13 [5.84]						0.60 [2.20]
Anomaly 2		0.21 [1.83]					0.10 [0.98]
Anomaly 3			0.25 [4.52]				0.25 [4.24]
Anomaly 4				0.38 [3.18]			-0.13 [-0.56]
Anomaly 5					0.12 [3.23]		0.17 [0.27]
Anomaly 6						0.13 [9.00]	0.80 [5.66]
# months	715	715	715	720	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21 [2.69]	0.24 [3.13]	0.16 [2.03]	0.21 [2.73]	0.22 [2.82]	0.21 [2.68]	0.16 [2.18]
Anomaly 1	24.19 [6.24]						10.47 [2.21]
Anomaly 2		12.77 [4.25]					4.31 [1.30]
Anomaly 3			25.55 [6.25]				13.69 [2.91]
Anomaly 4				25.58 [6.19]			31.20 [5.24]
Anomaly 5					7.20 [1.86]		-18.70 [-3.39]
Anomaly 6						6.60 [1.50]	2.15 [0.50]
mkt	0.80 [0.44]	2.20 [1.16]	1.35 [0.74]	1.07 [0.59]	0.10 [0.05]	0.26 [0.14]	3.10 [1.67]
smb	-0.19 [-0.07]	0.90 [0.33]	0.57 [0.21]	-1.98 [-0.75]	-1.22 [-0.45]	-1.06 [-0.39]	0.20 [0.08]
hml	-2.11 [-0.60]	-6.38 [-1.69]	0.56 [0.16]	-4.57 [-1.29]	-2.36 [-0.65]	-1.92 [-0.53]	-3.99 [-1.08]
rmw	-10.90 [-2.64]	-5.22 [-1.30]	-6.96 [-1.81]	3.45 [0.97]	2.90 [0.80]	2.30 [0.63]	-11.10 [-2.53]
cma	14.40 [2.53]	18.63 [3.21]	17.04 [3.09]	4.58 [0.70]	22.11 [3.36]	24.44 [3.93]	-0.28 [-0.04]
umd	2.19 [1.21]	4.31 [2.34]	1.62 [0.89]	2.85 [1.58]	3.49 [1.89]	3.18 [1.72]	1.20 [0.66]
# months	728	728	728	732	732	732	728
$\bar{R}^2(\%)$	12	10	12	12	8	8	17

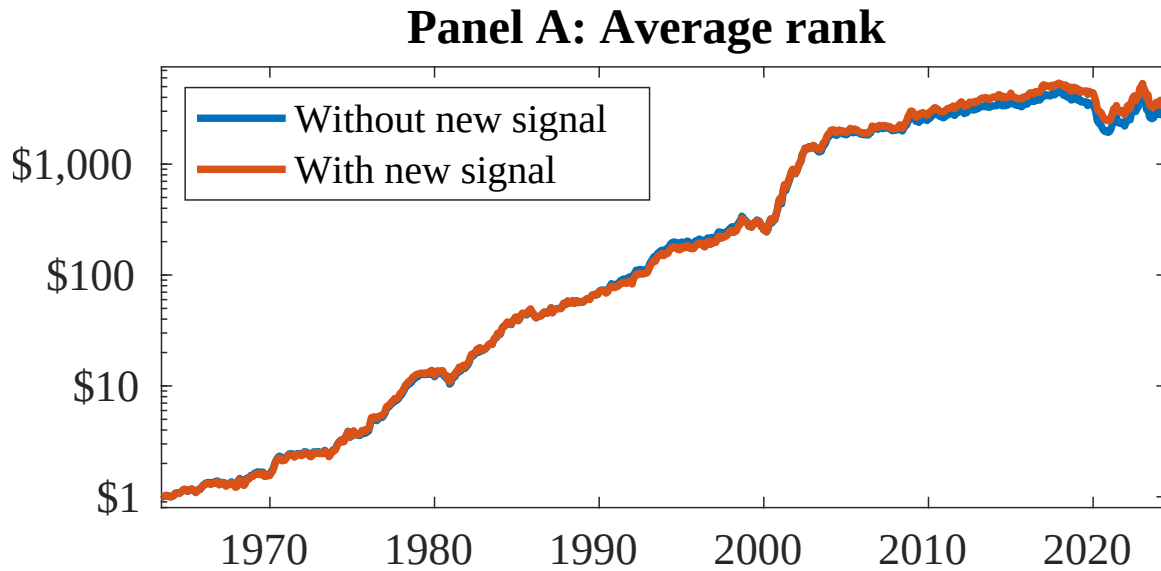


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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